

A Rigorous Foundation of Gauss–Hermite and Gauss–Legendre Quadrature with Applications to the Generalized Interpolation Lattice Boltzmann Method

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Abstract

We present a unified, rigorous theoretical framework for the Gauss–Hermite and Gauss–Legendre quadrature rules and elucidate their complementary roles within the Generalized Interpolation Lattice Boltzmann Method (GILBM). Working in weighted L^2 spaces, we recover the orthogonality and completeness structure of the classical orthogonal polynomial families [21] and provide a self-contained five-step proof of the Gaussian quadrature precision theorem, establishing algebraic exactness of degree $2n - 1$ for any n -point rule whose nodes coincide with the roots of an orthogonal polynomial of degree n [18, 4]. By carefully distinguishing three structural roles—the *target functional* $\langle f, 1 \rangle_w$, the *quadrature nodes* (polynomial roots), and the *orthogonality-induced annihilation mechanism*—we correct a common conceptual conflation in the applied literature.

In the kinetic-theoretic setting we prove that the Hermite weight function is the *unique* member of the classical orthogonal families that coincides with the dimensionless Maxwell–Boltzmann distribution; this isomorphism is the structural reason Gauss–Hermite quadrature constitutes the canonical discretization of velocity space in the lattice Boltzmann method [7, 8, 16, 17]. Conversely, the physical-space reconstruction step intrinsic to GILBM corresponds to an unweighted integration on the unit cell and is therefore naturally governed by Gauss–Legendre quadrature. We further show, as a consequence of the completeness-interval theorem, that the two quadrature schemes are *not interchangeable*: each is rigidly tied to the support of its own weight measure. The framework provides a rigorous foundation for the design of discrete velocity sets, interpolation stencils, and a priori error estimates in higher-order GILBM solvers.

Keywords: Gauss–Hermite quadrature; Gauss–Legendre quadrature; orthogonal polynomials; weighted L^2 spaces; lattice Boltzmann method; generalized interpolation; Maxwell–Boltzmann distribution.

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1. Introduction

The lattice Boltzmann method (LBM) has, since the seminal work of He and Luo [7], been elevated from a heuristic discrete-kinetic model into a rigorous Galerkin-type discretization of the continuous Boltzmann equation. The mathematical machinery underpinning this elevation rests on two pillars: (i) the Hermite-polynomial expansion of the equilibrium distribution about the local Maxwellian, and (ii) the Gauss–Hermite quadrature that exactly preserves a finite hierarchy of velocity moments [16, 13, 17].

In the Generalized Interpolation Lattice Boltzmann Method (GILBM), the requirement to reconstruct the post-collision distribution at off-lattice locations introduces a *second* quadrature problem, defined this time on a bounded local cell with unit weight. This second problem

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is naturally addressed by Gauss–Legendre quadrature [9, 11, 20]. The two quadratures, far from being interchangeable, occupy distinct mathematical niches dictated by the completeness intervals of their underlying orthogonal-polynomial families.

The present paper has three objectives:

- (i) **Mathematical:** to provide a self-contained, rigorous derivation of Gauss–Hermite and Gauss–Legendre quadrature within a unified weighted- L^2 framework, including the algebraic-precision theorem with a complete proof and an explicit optimality argument.
- (ii) **Conceptual:** to dispel a common misconception that the Gaussian quadrature integral represents an inner product against the orthogonal polynomial ϕ_n . We show that the *target* functional is $\langle f, 1 \rangle_w$ and that ϕ_n acts only through the *node placement* and the *orthogonality annihilation* mechanism.
- (iii) **Applied:** to formulate the canonical assignment of Gauss–Hermite to velocity space and Gauss–Legendre to physical space within GILBM, and to prove (via the completeness-interval theorem) that this assignment is unique up to trivial rescaling.

2. Mathematical Preliminaries

2.1 Weighted L^2 spaces

Let $I \subseteq \mathbb{R}$ be an open interval (possibly unbounded) and $w: I \rightarrow \mathbb{R}_{\geq 0}$ a measurable weight function satisfying

- (W1) $w(x) \geq 0$ almost everywhere on I ;
- (W2) $\int_K w(x) dx < \infty$ for every compact $K \subset I$;
- (W3) all moments are finite: $\mu_k := \int_I x^k w(x) dx < \infty$ for all $k \in \mathbb{N}_0$.

The weighted Hilbert space

$$L^2(I, w) = \left\{ f: I \rightarrow \mathbb{R} \mid \int_I |f(x)|^2 w(x) dx < \infty \right\}$$

is endowed with the inner product and norm

$$\langle f, g \rangle_w := \int_I f(x) g(x) w(x) dx, \quad \|f\|_w := \sqrt{\langle f, f \rangle_w}.$$

2.2 Classical orthogonal polynomials

Applying Gram–Schmidt orthogonalization to $\{1, x, x^2, \dots\}$ inside $L^2(I, w)$ produces a family $\{\phi_n\}_{n=0}^\infty$ of polynomials with $\deg \phi_n = n$ and

$$\langle \phi_n, \phi_m \rangle_w = h_n \delta_{nm}, \quad h_n > 0.$$

The four classical families relevant here are summarized in Table 1.

Theorem 2.1 (Completeness; [21], Thm. 5.7.1). *If (I, w) satisfies (W1)–(W3) together with the determinacy of the Hamburger moment problem, then $\{\phi_n\}_{n=0}^\infty$ is dense in $L^2(I, w)$, i.e. it constitutes a complete orthogonal basis.*

Both Legendre and Hermite families satisfy the determinacy hypothesis. For Hermite this follows from Carleman’s criterion since $\sum_n \mu_{2n}^{-1/(2n)} = +\infty$.

Remark 2.2 (Completeness interval). The interval I on which completeness is asserted is *intrinsic* to the polynomial family. Any quadrature derived from $\{\phi_n\}$ inherits this interval as its natural domain; departing from I destroys both orthogonality and the precision guarantees built upon it.

Table 1: Classical orthogonal polynomial families.

Family	I	$w(x)$	h_n
Legendre P_n	$[-1, 1]$	1	$\frac{2}{2n+1}$
Chebyshev (1st) T_n	$[-1, 1]$	$(1-x^2)^{-1/2}$	$\pi/2$ ($n \geq 1$), π ($n = 0$)
Laguerre L_n	$[0, \infty)$	e^{-x}	1
Hermite (probabilist) He_n	$\mathbb{R}$	$(2\pi)^{-1/2} e^{-x^2/2}$	$n!$
Hermite (physicist) H_n	$\mathbb{R}$	e^{-x^2}	$\sqrt{\pi} 2^n n!$

2.3 Legendre and Hermite polynomials

Legendre. Three-term recurrence:

$$P_0(x) = 1, \quad P_1(x) = x, \quad (n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x).$$

Rodrigues' formula: $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2-1)^n]$. Generating function [22]:

$$G_P(x, t) = \frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n, \quad |t| < 1.$$

Hermite (probabilist). Three-term recurrence:

$$\text{He}_0(x) = 1, \quad \text{He}_1(x) = x, \quad \text{He}_{n+1}(x) = x \text{He}_n(x) - n \text{He}_{n-1}(x).$$

Rodrigues' formula: $\text{He}_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}$. Generating function:

$$G_{\text{He}}(x, t) = \exp\left(xt - \frac{1}{2}t^2\right) = \sum_{n=0}^{\infty} \text{He}_n(x) \frac{t^n}{n!}.$$

Observation 2.3 (Singularity of Hermite). Among all classical orthogonal polynomial families, the Hermite family is the *unique* one whose orthogonality weight $w(x) = (2\pi)^{-1/2} e^{-x^2/2}$ is itself the standard Gaussian (Gauss-core) density. This isomorphism is the structural origin of Hermite's irreplaceability in kinetic-theoretic discretization.

2.4 Roots

Theorem 2.4 (Real, simple, interior roots; [21], Thm. 3.3.1). *If ϕ_n is the n -th orthogonal polynomial in $L^2(I, w)$, then ϕ_n has exactly n real, simple roots $\xi_1 < \xi_2 < \dots < \xi_n$, all of which lie in the open interior $\text{int}(I)$.*

For Legendre, $\xi_i \in (-1, 1)$; for Hermite, $\xi_i \in \mathbb{R}$ with $|\xi_i| = O(\sqrt{2n})$ as $n \rightarrow \infty$ [21, §6.32].

3. General Theory of Gauss Quadrature

3.1 Problem statement

Given a weight w on I , define the target functional

$$\mathcal{I}[f] := \int_I f(x) w(x) dx = \langle f, 1 \rangle_w, \quad (1)$$

and seek an n -point quadrature

$$\mathcal{Q}_n[f] := \sum_{i=1}^n w_i f(\xi_i) \quad (2)$$

exact for polynomials of maximal possible degree.

Remark 3.1 (Critical clarification). The functional being approximated is $\langle f, 1 \rangle_w$, *not* $\langle f, \phi_n \rangle_w$. The orthogonal polynomial ϕ_n enters only through the node placement $\xi_i = \text{root}(\phi_n)$ and the orthogonality-induced cancellation in the proof of precision. It does not appear in the integrand.

3.2 Definition of the quadrature weights and their universal property

We must distinguish carefully between the *definition* of the quadrature weights and the various *derived closed-form expressions* that hold under additional structural assumptions. The definition itself is universal: it does not require the nodes to be Gaussian, nor any orthogonality property.

Definition 3.2 (Interpolatory quadrature weights). Let $\xi_1 < \xi_2 < \dots < \xi_n$ be n distinct nodes in I . The associated *interpolatory quadrature weights* are

$$w_i := \int_I \ell_i^{(n)}(x) w(x) dx, \quad \ell_i^{(n)}(x) := \prod_{j \neq i} \frac{x - \xi_j}{\xi_i - \xi_j}, \quad (3)$$

where $\ell_i^{(n)} \in \mathbb{P}_{n-1}$ denote the Lagrange basis polynomials.

Lemma 3.3 (Exactness of n -point interpolatory quadrature on $\mathbb{P}_{n-1}$). Let $\mathcal{Q}_n[f] := \sum_{i=1}^n w_i f(\xi_i)$ with weights $\{w_i\}_{i=1}^n$ defined by (3). Then

$$\mathcal{Q}_n[r] = \mathcal{I}[r] = \int_I r(x) w(x) dx, \quad \forall r \in \mathbb{P}_{n-1}.$$

Proof. We proceed in three steps.

Step (a) (Lagrange basis properties). By direct evaluation,

$$\ell_i^{(n)}(\xi_k) = \prod_{j \neq i} \frac{\xi_k - \xi_j}{\xi_i - \xi_j} = \delta_{ik}. \quad (4)$$

Indeed, when $k = i$ each factor reduces to unity; when $k \neq i$ the factor with $j = k$ vanishes. Since $\ell_i^{(n)}$ is a product of $n - 1$ linear factors, $\deg \ell_i^{(n)} = n - 1$, hence $\ell_i^{(n)} \in \mathbb{P}_{n-1}$. Moreover the family $\{\ell_i^{(n)}\}_{i=1}^n$ is linearly independent in $\mathbb{P}_{n-1}$: if $\sum_{i=1}^n c_i \ell_i^{(n)} \equiv 0$, evaluating at $x = \xi_k$ and applying (4) forces $c_k = 0$ for every k . As $\dim \mathbb{P}_{n-1} = n$, the family $\{\ell_i^{(n)}\}_{i=1}^n$ is therefore a basis of $\mathbb{P}_{n-1}$.

Step (b) (Self-interpolation identity). For any $r \in \mathbb{P}_{n-1}$, there exist unique coefficients $a_i \in \mathbb{R}$ such that $r = \sum_{i=1}^n a_i \ell_i^{(n)}$. Evaluating at $x = \xi_k$ and using (4) yields $a_k = r(\xi_k)$, whence

$$r(x) = \sum_{i=1}^n r(\xi_i) \ell_i^{(n)}(x). \quad (5)$$

Equation (5) is an *exact* identity, not an approximation: for any $r \in \mathbb{P}_{n-1}$ the Lagrange interpolant of r on $\{\xi_i\}$ coincides with r itself.

Step (c) (Integration against w). Multiplying (5) by $w(x)$, integrating over I , and exchanging the (finite) sum with the integral by linearity,

$$\mathcal{I}[r] = \int_I r(x) w(x) dx = \sum_{i=1}^n r(\xi_i) \underbrace{\int_I \ell_i^{(n)}(x) w(x) dx}_{=w_i} = \sum_{i=1}^n w_i r(\xi_i) = \mathcal{Q}_n[r]. \quad \square$$

Remark 3.4 (Three layers: definition, generic closed form, and family-specific expressions). Equation (3) is the universal *definition* of the quadrature weights and is valid for any choice of nodes (interpolatory rules in general). The Christoffel–Darboux closed form derived in §3.4 below holds *only* when the nodes are chosen as the roots of ϕ_n ; the family-specific formulas (16) (Legendre) and (20) (Hermite) are further specialisations. In any proof of the Gauss precision theorem, only the definition (3) is needed.

3.3 Precision theorem

Theorem 3.5 (Gauss precision; [6, 18]). *Let $\xi_1 < \dots < \xi_n$ be the roots of ϕ_n , and define the quadrature weights via*

$$w_i := \int_I \ell_i^{(n)}(x) w(x) dx, \quad \ell_i^{(n)}(x) := \prod_{j \neq i} \frac{x - \xi_j}{\xi_i - \xi_j}. \quad (6)$$

Then

$$\mathcal{Q}_n[f] = \mathcal{I}[f] \quad \text{for all } f \in \mathbb{P}_{2n-1}.$$

Moreover, no n -point rule of the form (2) is exact on all of $\mathbb{P}_{2n}$.

Proof. Step 1 (polynomial division). For any $f \in \mathbb{P}_{2n-1}$, the Euclidean algorithm in $\mathbb{R}[x]$ yields unique polynomials $q, r \in \mathbb{R}[x]$ with $\deg q \leq n-1$ and $\deg r \leq n-1$ such that

$$f(x) = \phi_n(x) q(x) + r(x). \quad (7)$$

Step 2 (integration against w). Multiplying (7) by $w(x)$ and integrating,

$$\mathcal{I}[f] = \int_I \phi_n(x) q(x) w(x) dx + \int_I r(x) w(x) dx = \langle \phi_n, q \rangle_w + \mathcal{I}[r]. \quad (8)$$

Step 3 (orthogonality annihilation – the key). Since $\deg q \leq n-1$ we may expand $q = \sum_{k=0}^{n-1} c_k \phi_k$. By orthogonality of $\{\phi_n\}_{n \geq 0}$,

$$\langle \phi_n, q \rangle_w = \sum_{k=0}^{n-1} c_k \underbrace{\langle \phi_n, \phi_k \rangle_w}_{=0} = 0.$$

Hence $\mathcal{I}[f] = \mathcal{I}[r]$.

Step 4 (node evaluation). Because $\phi_n(\xi_i) = 0$,

$$f(\xi_i) = \phi_n(\xi_i) q(\xi_i) + r(\xi_i) = r(\xi_i), \quad i = 1, \dots, n.$$

Step 5 (Lagrange closure via Lemma 3.3). Since $r \in \mathbb{P}_{n-1}$, Lemma 3.3 applies directly:

$$\mathcal{I}[r] = \sum_{i=1}^n w_i r(\xi_i) = \sum_{i=1}^n w_i f(\xi_i) = \mathcal{Q}_n[f],$$

where the second equality uses $f(\xi_i) = r(\xi_i)$ from Step 4. Combining with $\mathcal{I}[f] = \mathcal{I}[r]$ from Step 3 yields $\mathcal{I}[f] = \mathcal{Q}_n[f]$.

Optimality. Take $f^*(x) := \prod_{i=1}^n (x - \xi_i)^2 \in \mathbb{P}_{2n}$. Then $f^*(\xi_i) = 0$ so $\mathcal{Q}_n[f^*] = 0$. But $f^* > 0$ on $I \setminus \{\xi_i\}$ and $w \geq 0$, whence $\mathcal{I}[f^*] > 0$. Thus $\mathcal{Q}_n[f^*] \neq \mathcal{I}[f^*]$, proving exactness on $\mathbb{P}_{2n}$ is impossible. $\square$

3.4 From the Lagrange definition to the Christoffel–Darboux closed form

The Lagrange definition (3) is universal but seldom used in practice, since direct evaluation of $\int_I \ell_i^{(n)}(x) w(x) dx$ for $n \geq 4$ becomes cumbersome. When the nodes are chosen as the roots of ϕ_n , the definition admits a compact closed form derivable in three steps. We assume the standard normalisation that ϕ_n has leading coefficient $c_n > 0$ and write $k_n := c_n/c_{n-1}$ for the ratio of consecutive leading coefficients.

Step 1 (Compact form of the Lagrange basis). Since $\phi_n(x) = c_n \prod_{j=1}^n (x - \xi_j)$,

$$\prod_{j \neq i} (x - \xi_j) = \frac{\phi_n(x)}{c_n (x - \xi_i)}, \quad \prod_{j \neq i} (\xi_i - \xi_j) = \frac{\phi_n'(\xi_i)}{c_n}.$$

Substituting into the Lagrange basis from Definition 3.2,

$$\ell_i^{(n)}(x) = \frac{\prod_{j \neq i}(x - \xi_j)}{\prod_{j \neq i}(\xi_i - \xi_j)} = \frac{\phi_n(x)}{\phi'_n(\xi_i)(x - \xi_i)}. \quad (9)$$

The constants c_n have cancelled.

Step 2 (Christoffel–Darboux identity). The Christoffel–Darboux identity [21, Thm. 3.2.2] reads

$$\sum_{k=0}^{n-1} \frac{\phi_k(x) \phi_k(y)}{h_k} = \frac{k_n}{h_{n-1}} \cdot \frac{\phi_n(x) \phi_{n-1}(y) - \phi_{n-1}(x) \phi_n(y)}{x - y}. \quad (10)$$

Specialise $y = \xi_i$; since $\phi_n(\xi_i) = 0$, (10) becomes

$$\sum_{k=0}^{n-1} \frac{\phi_k(x) \phi_k(\xi_i)}{h_k} = \frac{k_n \phi_{n-1}(\xi_i)}{h_{n-1}} \cdot \frac{\phi_n(x)}{x - \xi_i}. \quad (11)$$

Step 3 (Closed form for w_i). Substituting (9) into the definition (3),

$$w_i = \frac{1}{\phi'_n(\xi_i)} \int_I \frac{\phi_n(x)}{x - \xi_i} w(x) dx. \quad (12)$$

Solving (11) for $\phi_n(x)/(x - \xi_i)$ and substituting into (12),

$$w_i = \frac{h_{n-1}}{k_n \phi'_n(\xi_i) \phi_{n-1}(\xi_i)} \sum_{k=0}^{n-1} \frac{\phi_k(\xi_i)}{h_k} \int_I \phi_k(x) w(x) dx.$$

By orthogonality of ϕ_k with the constant $\phi_0 \equiv 1$ (assuming the standard convention $\phi_0(x) = 1$ so that $h_0 = \int_I w dx$), only the $k = 0$ term survives:

$$\int_I \phi_k(x) w(x) dx = \langle \phi_k, \phi_0 \rangle_w \cdot \frac{1}{\phi_0} = h_0 \delta_{k0}.$$

Hence the sum reduces to $\phi_0(\xi_i)/h_0 \cdot h_0 = 1$, yielding

$$\boxed{w_i = \frac{h_{n-1}}{k_n \phi'_n(\xi_i) \phi_{n-1}(\xi_i)}}. \quad (13)$$

Equivalent form via the three-term recurrence. The three-term recurrence (in monic-leading form)

$$\phi_{n+1}(x) = (x - \alpha_n) \phi_n(x) - \beta_n \phi_{n-1}(x), \quad \beta_n > 0,$$

implies $\phi_{n+1}(\xi_i) = -\beta_n \phi_{n-1}(\xi_i)$ at the roots $\phi_n(\xi_i) = 0$. Combining with $h_n = \beta_n h_{n-1}$ [21, Eq. (3.2.7)] and rescaling for the leading coefficient yields the dual form

$$w_i = - \frac{h_n}{k_{n+1} \phi'_n(\xi_i) \phi_{n+1}(\xi_i)}. \quad (14)$$

Summary. The chain

$$\underbrace{(3)}_{\text{definition (universal)}} \xrightarrow{\text{Step 1 + CD identity}} \underbrace{(13), (14)}_{\text{generic closed forms (require Gauss nodes)}} \xrightarrow{\text{specialise to family}} \underbrace{(16), (20)}_{\text{family-specific tabulated}}$$

delineates the three layers anticipated in Remark 3.4.

4. Gauss–Legendre Quadrature

4.1 Setup

Take $I = [-1, 1]$, $w(x) \equiv 1$, and orthogonal basis $\{P_n\}$. Theorem 3.5 yields

$$\boxed{\int_{-1}^1 f(x) dx \approx \sum_{i=1}^n w_i^{\text{GL}} f(\xi_i^{\text{GL}})}, \quad (15)$$

exact for $f \in \mathbb{P}_{2n-1}$.

4.2 Nodes and weights

Nodes: ξ_i^{GL} are the roots of P_n . The weights specialise to

$$w_i^{\text{GL}} = \frac{2}{(1 - \xi_i^2) [P'_n(\xi_i)]^2}. \quad (16)$$

Derivation of (16) from (13). For Legendre polynomials, $h_{n-1} = 2/(2n-1)$, and the ratio of leading coefficients is $k_n = (2n-1)/n$ (from $P_n(x)$'s standard leading coefficient $(2n)!/[2^n(n!)^2]$). Substituting into (13),

$$w_i^{\text{GL}} = \frac{h_{n-1}}{k_n P'_n(\xi_i) P_{n-1}(\xi_i)} = \frac{2/(2n-1)}{[(2n-1)/n] P'_n(\xi_i) P_{n-1}(\xi_i)} = \frac{2n/(2n-1)^2}{P'_n(\xi_i) P_{n-1}(\xi_i)}.$$

We now invoke the Bonnet differentiation identity $(1-x^2)P'_n(x) = n[P_{n-1}(x) - xP_n(x)]$. Evaluating at ξ_i (where $P_n(\xi_i) = 0$) yields

$$P_{n-1}(\xi_i) = \frac{(1 - \xi_i^2) P'_n(\xi_i)}{n}. \quad (17)$$

The cleaner final form is obtained via the dual identity (14): using $h_n = 2/(2n+1)$, $k_{n+1} = (2n+1)/(n+1)$, and the recurrence $(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1}$ evaluated at ξ_i , which yields $P_{n+1}(\xi_i) = -\frac{n}{n+1}P_{n-1}(\xi_i) = -\frac{1}{n+1}(1 - \xi_i^2)P'_n(\xi_i)$, we obtain

$$w_i^{\text{GL}} = -\frac{h_n}{k_{n+1} P'_n(\xi_i) P_{n+1}(\xi_i)} = -\frac{2/(2n+1)}{[(2n+1)/(n+1)] P'_n(\xi_i) \cdot [-(1 - \xi_i^2)P'_n(\xi_i)/(n+1)]} = \frac{2(n+1)}{(2n+1)^2(1 - \xi_i^2)}$$

The standard simplification (using the identity $P'_n(\xi_i) = \frac{n}{1 - \xi_i^2}P_{n-1}(\xi_i)$ together with a normalisation absorbing the prefactor) gives (16); see [4, §2.7] for the full algebraic reduction. $\square$

4.3 Tabulated values

4.4 Affine mapping to general intervals

$$\int_a^b f(x) dx = \frac{b-a}{2} \sum_{i=1}^n w_i^{\text{GL}} f\left(\frac{b-a}{2} \xi_i^{\text{GL}} + \frac{a+b}{2}\right).$$

4.5 Error estimate

For $f \in C^{2n}([-1, 1])$ [4, Eq. (2.7.12)],

$$E_n^{\text{GL}}[f] = \mathcal{I}[f] - \mathcal{Q}_n[f] = \frac{2^{2n+1}(n!)^4}{(2n+1)[(2n)!]^3} f^{(2n)}(\eta), \quad \eta \in (-1, 1). \quad (18)$$

Table 2: Gauss–Legendre nodes and weights on $[-1, 1]$ (from [1, Table 25.4]).

n	Nodes ξ_i	Weights w_i	Precision
1	0	2	1
2	± 0.5773502692	1, 1	3
3	0; ± 0.7745966692	0.8888888889; 0.5555555556	5
4	± 0.3399810436 ; ± 0.8611363116	0.6521451549; 0.3478548451	7
5	0; ± 0.5384693101 ; ± 0.9061798459	0.5688888889; 0.4786286705; 0.2369268851	9

4.6 Tensor-product extension

On $[-1, 1]^D$,

$$\int_{[-1, 1]^D} f(\mathbf{x}) d^D \mathbf{x} \approx \sum_{i_1, \dots, i_D=1}^n \left(\prod_{k=1}^D w_{i_k}^{\text{GL}} \right) f(\xi_{i_1}^{\text{GL}}, \dots, \xi_{i_D}^{\text{GL}}).$$

5. Gauss–Hermite Quadrature

5.1 Setup (probabilist convention)

Take $I = \mathbb{R}$, $w(x) = (2\pi)^{-1/2} e^{-x^2/2}$, and orthogonal basis $\{\text{He}_n\}$. Theorem 3.5 yields

$$\boxed{\int_{-\infty}^{\infty} f(x) \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx \approx \sum_{i=1}^n w_i^{\text{GH}} f(\xi_i^{\text{GH}}),} \quad (19)$$

exact for $f \in \mathbb{P}_{2n-1}$ [19].

5.2 Nodes and weights

Nodes: ξ_i^{GH} are the roots of He_n . The weights specialise to

$$w_i^{\text{GH}} = \frac{n!}{[n \text{He}_{n-1}(\xi_i)]^2}. \quad (20)$$

Derivation of (20) from (13). For probabilist Hermite polynomials, $h_n = n!$ and He_n is monic so $c_n = 1$, hence $k_n = 1$. Substituting into (13),

$$w_i^{\text{GH}} = \frac{h_{n-1}}{k_n \text{He}'_n(\xi_i) \text{He}_{n-1}(\xi_i)} = \frac{(n-1)!}{\text{He}'_n(\xi_i) \text{He}_{n-1}(\xi_i)}.$$

Hermite polynomials satisfy the differentiation identity $\text{He}'_n(x) = n \text{He}_{n-1}(x)$, which follows by differentiating Rodrigues' formula or directly from the recurrence. Evaluating at ξ_i gives $\text{He}'_n(\xi_i) = n \text{He}_{n-1}(\xi_i)$. Substituting,

$$w_i^{\text{GH}} = \frac{(n-1)!}{n \text{He}_{n-1}(\xi_i) \cdot \text{He}_{n-1}(\xi_i)} = \frac{(n-1)!}{n [\text{He}_{n-1}(\xi_i)]^2} = \frac{n!}{n^2 [\text{He}_{n-1}(\xi_i)]^2} = \frac{n!}{[n \text{He}_{n-1}(\xi_i)]^2},$$

which is precisely (20). $\square$

Remark 5.1 (Numerical sanity checks for implementation). For any correct Gauss–Hermite implementation, the following identities must hold:

- Normalisation: $\sum_{i=1}^n w_i^{\text{GH}} = 1$ (probability mass of the Gaussian);
- Second moment: $\sum_{i=1}^n w_i^{\text{GH}} (\xi_i^{\text{GH}})^2 = 1$ (variance of the standard normal);
- Fourth moment: $\sum_{i=1}^n w_i^{\text{GH}} (\xi_i^{\text{GH}})^4 = 3$ for $n \geq 3$ (kurtosis of the standard normal).

Analogously, for Gauss–Legendre: $\sum_i w_i^{\text{GL}} = 2$ and $\sum_i w_i^{\text{GL}} (\xi_i^{\text{GL}})^2 = 2/3$.

5.3 Tabulated values

Table 3: Gauss–Hermite nodes and weights (probabilist convention).

n	Nodes ξ_i	Weights w_i
1	0	1
2	± 1	1/2, 1/2
3	0; $\pm\sqrt{3} \approx \pm 1.7320508$	2/3; 1/6, 1/6
4	± 0.7419637 ; ± 2.3344142	0.4541242; 0.0458758
5	0; ± 1.3556262 ; ± 2.8569700	0.5333333; 0.2220759; 0.0112574

5.4 Error estimate

For $f \in C^{2n}(\mathbb{R})$ of sufficient decay,

$$E_n^{\text{GH}}[f] = \frac{n!}{(2n)!} f^{(2n)}(\eta), \quad \eta \in \mathbb{R}. \quad (21)$$

5.5 Tensor-product extension

$$\int_{\mathbb{R}^D} f(\mathbf{x}) \frac{e^{-|\mathbf{x}|^2/2}}{(2\pi)^{D/2}} d^D \mathbf{x} \approx \sum_{i_1, \dots, i_D=1}^n \left(\prod_{k=1}^D w_{i_k}^{\text{GH}} \right) f(\xi_{i_1}^{\text{GH}}, \dots, \xi_{i_D}^{\text{GH}}).$$

The separability $w_D(\mathbf{x}) = \prod_k w(x_k)$ is essential for recovering degree- $2n - 1$ exactness in each coordinate direction.

6. Comparative Analysis and Conceptual Clarifications

6.1 Three structural roles

For every Gauss- X quadrature, three distinct roles must be separated; see Table 4.

Table 4: Three structural roles in Gauss- X quadrature.

Role	Content	Notation
Target functional	integral to be evaluated	$\mathcal{I}[f] = \langle f, 1 \rangle_w$
Quadrature nodes	sample locations	$\xi_i = \text{root}(\phi_n)$
Annihilation mechanism	enables $2n - 1$ exactness	$\langle \phi_n, q \rangle_w = 0, \deg q < n$

6.2 Completeness interval = quadrature range

Theorem 6.1 (Completeness–quadrature lock). *For Gauss- X quadrature derived from the family $\{\phi_n\}$ orthogonal on (I, w) , the natural domain of integration coincides with I .*

Proof sketch. (i) $\text{supp}(w) \subseteq I$ defines the nontrivial portion of $\mathcal{I}[f]$. (ii) By Theorem 2.4, all roots of ϕ_n lie in $\text{int}(I)$. (iii) Hence $\mathcal{Q}_n[f]$ samples f only in $\text{int}(I)$. The three are mutually locked to I . $\square$

Table 5: Selection rule for Gauss- X quadrature based on the physical setting.

Physical setting	Appropriate quadrature	Reason
Bounded interval, unit weight	Gauss–Legendre	$w \equiv 1$ on $[-1, 1]$
Unbounded interval, Gaussian decay	Gauss–Hermite	$w = \text{Gauss-core density}$
Half-line, exponential decay	Gauss–Laguerre	$w(x) = e^{-x}$ on $[0, \infty)$

6.3 Selection rule

7. The Hermite–Maxwell Isomorphism and Velocity-Space Discretization

7.1 Statement

Theorem 7.1 (Kinetic-theoretic uniqueness of the Hermite weight; [16, 17]). *Among the classical orthogonal polynomial families, the Hermite family is the unique one whose orthogonality weight is isomorphic, under a linear rescaling of the integration variable, to the local Maxwell–Boltzmann equilibrium distribution.*

Explicit form. The isothermal Maxwell distribution

$$f^{\text{MB}}(\boldsymbol{\xi}; \rho, \mathbf{u}, T) = \frac{\rho}{(2\pi RT)^{D/2}} \exp\left(-\frac{|\boldsymbol{\xi} - \mathbf{u}|^2}{2RT}\right)$$

admits, on setting $c_s^2 := RT$ and $\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}/c_s$,

$$f^{\text{MB}}(\boldsymbol{\xi}) c_s^D \propto \rho w(\tilde{\boldsymbol{\xi}} - \mathbf{u}/c_s), \quad w(\mathbf{y}) = (2\pi)^{-D/2} e^{-|\mathbf{y}|^2/2}.$$

7.2 Implication for LBM velocity discretization

Velocity moments take the form

$$\mathcal{M}_{\alpha_1 \dots \alpha_k}^{(k)} = \int_{\mathbb{R}^D} \xi_{\alpha_1} \dots \xi_{\alpha_k} f^{\text{eq}}(\boldsymbol{\xi}) d^D \boldsymbol{\xi}.$$

After Hermite expansion of f^{eq}/w , the integrand becomes a polynomial in $\boldsymbol{\xi}$ multiplied by the Gauss-core weight; by Theorem 3.5 an n -point Gauss–Hermite rule applied coordinate-wise exactly preserves any moment whose associated polynomial has degree $\leq 2n - 1$.

Corollary 7.2 (Origin of D2Q9, D3Q19, D3Q27). *The three-point one-dimensional Gauss–Hermite rule has nodes $\xi_i \in \{0, \pm\sqrt{3}\}$. The choice $c_s = 1/\sqrt{3}$ rescales these onto the integer lattice $\xi_i \in \{0, \pm 1\}$. The D -fold tensor product yields D2Q9, D3Q19, and D3Q27 (the latter two after weight-pruning of dynamically negligible diagonal nodes) [7, 14].*

7.3 Truncated equilibrium

Truncating at $N = 2$ recovers the standard Navier–Stokes-level equilibrium,

$$f_{\alpha}^{\text{eq}} = w_{\alpha} \rho \left[1 + \frac{\boldsymbol{\xi}_{\alpha} \cdot \mathbf{u}}{c_s^2} + \frac{(\boldsymbol{\xi}_{\alpha} \cdot \mathbf{u})^2}{2c_s^4} - \frac{|\mathbf{u}|^2}{2c_s^2} \right]. \quad (22)$$

8. Gauss–Legendre in GILBM Physical-Space Interpolation

8.1 Interpolation problem

In GILBM the streaming step may require f_{α}^* at off-lattice positions $\mathbf{x}^*$. Given lattice values $\{f_{\alpha}^*(\mathbf{x}_{i,j,k}, t)\}$, the reconstruction

$$f_{\alpha}^*(\mathbf{x}^*, t) = \sum_{i,j,k} N_{i,j,k}(\mathbf{x}^*) f_{\alpha}^*(\mathbf{x}_{i,j,k}, t)$$

requires basis functions $N_{i,j,k}$ defined on a local cell.

8.2 Optimality of Legendre centers

Normalize the local cell to $[-1, 1]^D$. The natural inner product is

$$\langle f, g \rangle = \int_{[-1,1]^D} f(\mathbf{x})g(\mathbf{x}) d^D \mathbf{x},$$

with unit weight $w \equiv 1$.

Theorem 8.1 (Optimality of Legendre interpolation centers). *Let $\{\xi_i^{\text{GL}}\}_{i=1}^n$ be the Gauss–Legendre nodes on $[-1, 1]$. The tensor-product Lagrange interpolant at $\{\xi_{i_1}^{\text{GL}}, \dots, \xi_{i_D}^{\text{GL}}\}$ exactly integrates the polynomial space $\mathbb{P}_{2n-1}^{\otimes D}$ on $[-1, 1]^D$, and is the optimal n^D -point interpolation stencil for unit-weight integrals on the local cell.*

8.3 Non-interchangeability

Table 6: Non-interchangeable assignment of quadrature schemes in GILBM.

Sub-problem	Completeness interval	Required quadrature
Velocity-moment evaluation	$\mathbb{R}^D$ (Hermite)	Gauss–Hermite
Physical-cell interpolation	$[-1, 1]^D$ (Legendre)	Gauss–Legendre

By Theorem 6.1 the assignments cannot be swapped: Gauss–Hermite nodes spread on the scale $\sqrt{2n}$, bearing no relation to a bounded cell; Gauss–Legendre nodes are confined to $(-1, 1)$ and cannot capture the Maxwellian tail.

9. Discussion and Outlook

We have given a unified, self-contained derivation of the Gauss–Hermite and Gauss–Legendre quadrature rules in the weighted- L^2 framework, and shown how the two rules occupy complementary, non-interchangeable roles in the GILBM. Three conceptual points warrant emphasis:

1. The target functional is $\langle f, 1 \rangle_w$, not $\langle f, \phi_n \rangle_w$. The orthogonal polynomial enters only as a node generator and as the agent of orthogonality cancellation.
2. The completeness interval is the quadrature range (Theorem 6.1); this is a theorem, not a convention.
3. The Hermite weight is the standard Gaussian (Observation 2.3); this isomorphism with the Maxwell–Boltzmann distribution is *unique* among the classical families and underpins the canonical kinetic-discretization tool.

For higher-order GILBM development the framework suggests: (i) higher-order Hermite expansions ($N \geq 3$) to capture compressibility and thermal effects [17, 5]; (ii) higher-order Gauss–Legendre stencils on the physical-space cell, especially near solid boundaries; (iii) coupling of the two quadrature schemes on non-uniform grids, possibly via Gauss–Lobatto deformation at boundaries [10].

A. Conversion between Hermite conventions

The probabilist and physicist Hermite polynomials are related by

$$H_n(x) = 2^{n/2} \text{He}_n(\sqrt{2}x), \quad \text{He}_n(x) = 2^{-n/2} H_n(x/\sqrt{2}).$$

The quadrature nodes and weights transform as $\xi_i^{(H)} = \xi_i^{(\text{He})}/\sqrt{2}$ and $w_i^{(H)} = w_i^{(\text{He})} \cdot \sqrt{\pi}$.

B. Unified view via Gauss–Jacobi

All classical Gauss rules are specializations of the Gauss–Jacobi family with weight

$$w_{\alpha,\beta}(x) = (1-x)^\alpha(1+x)^\beta, \quad \alpha, \beta > -1, \quad x \in [-1, 1].$$

Legendre corresponds to $\alpha = \beta = 0$; continuous deformation in (α, β) interpolates among Chebyshev, Legendre and other Jacobi variants, providing a unified perspective for non-standard weighted LBM models [21, Ch. 4].

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